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LET'S TALK NLP AND LLM!

Focus for today

- What is NLP? What do we mean by LLM?
- NLP Basic
- NLP and sentiment analysis
- A case study

NLP & LLM

Natural Language Processing (NLP)

- NLP is the application of tasks that understand the human language: speech recognition, translation, finding synonyms
- The field is experiencing a rapid acceleration in applications thanks to the development of Large Language Model (LLM)- e.g. Google BERT, OpenAI GPT3
- LLMs are deep learning algorithm and so defined because of the large number of parameters (up to billions); they learn semantic and syntactic relationships among words from a large volume of text and this enables them to take into account context-major step advances relative traditional BoW methods that rely purely on frequency of words.
- Key applications in finance include sentiment analysis, topic modelling, entity recognition, text summarization

NLP Basics

NLP Basics

1. Sentence or Word tokenization: the process of tokenizing or splitting a string , text into a list of tokens e.g. a word is a token in a sentence, a sentence is a token in a paragraph.

2. Text lemmatization and stemming: both techniques attempt to map multiple words to a common root. Stemming cuts off the end and the beginning of the word taking into account a list of common prefixes and suffixes (e.g. “studying”, suffix “ing”; stem=study); lemmatization takes into consideration the morphological analysis of the words/requires detailed dictionaries to link the word back to its lemma

3. Stop Words: usually words like articles and pronouns; they get removed as part of pre-processing; increases algorithm performance (e.g. searches)

4. Sparse Vectors Challenge: A sparse vector is a vector having a relatively small number of non-zero elements; memory and computational challenges (zero elements with no info still take up memory and eats computational time)

5. BoW & TD-IDF

NLP continued: BOW & TF-IDF

- Bag of words model is a simplifying representation used in NLP; approach based on the frequency /occurrence of a word
- Word frequency in a document or a body of documents; no reference to context(grammar, order etc)
- Term Frequency-Inverse Document Frequency (TF-IDF)
  - The more frequently a word appears in a document, the more its importance increases; the more documents a word appears in, the more its importance decreases
  - Intuition: downplay words like definite or indefinite articles
  - Higher complexity

Source: https://quantdare.com/encoding-financial-texts-into-dense-representations/

NLP continued: TF-IDF

$$w_{ij} = tf_{ij} * log\frac{N}{df_i}$$

Where:

- w_{ij} is the TF-IDF of word *i* in document *j*
- tf_{ij} is the number of occurrences of word *i* in document *j*
- *N* is the number of documents
- df_i is the number of documents containing word *i* at least once

Source: https://quantdare.com/encoding-financial-texts-into-dense-representations/

NLP & Sentiment Analysis

NLP and sentiment indicators

Sentiment indicators in finance:

- Measures of how a group (e.g. investors, businesses, experts) perceives the state of markets/economy/sectors/specific company
- Survey based sentiment indicators of the economy (University of Michigan and Conference Board)
- In finance, often used as contrarian indicator that work best at extreme readings (e.g. bull/bear ratios)

NLP continued: sentiment analysis

- Sentiment analysis is a well-known research area in computational text analysis.

- Key considerations:

  - Domain specificity: this is very relevant in financial modelling as words may have different connotations than in every-day parlance; preference for use of LM lexicon vs GI.
  - Complexity: composition/context

- Lexical methodology where words are pre-assigned a sentiment score , usually 1,0,-1

NLP Continued: lexicon options

Table 1: Size and Coverage of Lexicons

			Unique Words		News Coverage Rate	
Lexicon	Feature Space	Size	Number	Fraction	Unique	Total
General Inquirer (GI)	General English	3626	1117	0.308	0.037	0.064
Loughran-McDonald (LM)	10-K Reports	2707	1569	0.580	0.007	0.028
Hu-Liu (HL)	Movie Reviews	6786	3832	0.565	0.013	0.044
Combined (GI+LM+HL)		9570				0.091

Notes: The total news coverage rate is the fraction of unigrams in the news corpus that are scored by the lexicon. The unique news coverage rate represents the fraction of unigrams in the news corpus that are scored only by the lexicon among the three.

Source: Shapiro, Adam Hale, Moritz Sudhof, and Daniel Wilson. 2017. "Measuring News Sentiment," Federal Reserve Bank of San Francisco Working Paper 2017-01

NLP continued: sentiment analysis

- Enhanced lexical methodology: Vader is a sentence-level sentiment classifier. It combines a lexicon which classifies words from +4 to -4 (most positives to most negative) and set of heuristic rules to help contextualise; Vader has been developed for social media and its efficacy in finance is more questionable.

NLP (continued): Vader heuristic rules

- Punctuation — I love pizza vs I love pizza!!
- Capitalization — I'm hungry!! vs I'M HUNGRY!!
- Degree modifiers (use of intensifiers) — I WANT TO EAT!! VS I REALLY WANT TO EAT!!
- Conjunctions (shift in sentiment polarity, with later dictating polarity) — I love pizza, but I really hate Pizza Hut (bad review)
- Preceding Tri-gram (identifying reverse polarity by examining the tri-gram before the lexical feature) — Canadian Pizza is not really *all that great*.

Source: https://towardsdatascience.com/https-towardsdatascience-com-algorithmic-trading-using-sentiment-analysis-on-news-articles-83db77966704

NLP Continued: Sources of data

Impossible to draw an exhaustive list, but a few to be familiar with:

- Common Crawl(https://commoncrawl.org/) a free open repository of web crawls that can be used by anyone
- Wikipedia (https://en.wikipedia.org/wiki/Main_Page) the free encyclopaedia that anyone can use
- NYT
- Yahoo
- BBC news

NLP & Finance: A Case Study

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Review and discussion of the paper "Measuring News Sentiment"

Shapiro, Adam Hale, Moritz Sudhof, and Daniel Wilson. 2017. "Measuring News Sentiment," Federal Reserve Bank of San Francisco Working Paper 2017-01

Key learnings

1. Very clear purpose and thesis: Assess predictive accuracy of sentiment analysis NLP based models, hence their potential as substitutes to traditional survey based measures.

2. Size of universe: 231k articles from 16 papers spanning period 01/1980 to 04/2015. Apply key filters:

    1. LN classified “country subject” as “United States” (with an LN “relevance” threshold of at least 85%).
    2. LN classified “topic subject” as “Economy” or “Economic” (with an LN “relevance” threshold of at least 85%).
    3. LN did NOT classify the article as a “Brief” or “Summary” or “Digest.“
    4. Article had 200 words or longer.
    5. Article contained at least one of the following words: said, says, told, stated, wrote, reported.

Key learnings

3. Assess coverage of your lexicon-Table 1

4. Define a benchmark of what good looks like-800 articles rated by Fed researchers- “human benchmark”

5. Assess how the various lexicon options perform relative to the human benchmark. Key techniques: Spearman rank correlation, R2 from an OLS regression of the human ratings on the model scores, the pseudo-R2 from an ordered logit regression, Macro-F1 statistic of classification accuracy

Key learnings

Table 2: Goodness-of-Fit of Lexical Model Sentiment Scores for Predicting Human Ratings

Model	Feature Space	Lexicon Size	Ordered-Logit Pseudo R^2	OLS R^2	Rank Correlation	Mac
GI Lexicon	General English	3626	0.023	0.064	0.264	0.
+ Negation Rule			0.029	0.080	0.295	0.
LM Lexicon	10-K Reports	2707	0.065	0.165	0.447	0.
+ Negation Rule			0.066	0.169	0.449	0.
HL Lexicon	Movie Reviews	6786	0.066	0.173	0.437	0.
+ Negation Rule			0.072	0.186	0.453	0.
GI + LM + HL Lexicon	Combined	9570	0.063	0.163	0.426	0.
+ Negation Rule			0.070	0.180	0.444	0.
LM + HL Lexicon	Combined	8453	0.077	0.195	0.476	0.
+ Negation Rule			0.081	0.205	0.486	0.
Vader	Social Media	7502	0.068	0.168	0.438	0.
Vader (incl. LM + HL lexicons)	Combined	9057	0.075	0.189	0.452	0.
News Lexicon	Econ/Finance	50754	0.071	0.180	0.459	0.
+ Negation Rule	News Articles		0.075	0.188	0.469	0.
News Lexicon + LM + HL	Combined	50754	0.078	0.197	0.480	0.
+ Negation Rule			**0.082**	**0.206**	**0.491**	**0.**

Notes: GI, LM, and HL refer, respectively, to the following lexicons: Harvard General Inquirer; Loughran and McDonald(2011), updated in 2014; and Hu and Liu (2004). The goodness-of-fit statistics are calculated using the full 800-article sample for which we have human ratings.

Key learnings

6. How to enhance the lexicon to deal with “unclassified” words

- Assign a sentiment class to each sentence using modified Vader
- Create a word-by-class matrix, counting the co-occurrence of each word with each of the sentiment classes.
- Estimate the implied sentiment association of each word via PMI (“pointwise mutual information”)

7. Build an index based on the new news sentiment metric and measure correlation with established surveys of economic activity

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